Project Plan: Warehouse Management System Replacement

Northwind Freight (fictional scenario)

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Subject	43030 Professional Practice in Computing, Assessment Task 3
Purpose	Goal 4 evidence: selecting and applying a project delivery methodology
Note	Northwind Freight is a fictional company. All names, figures and dates in this plan are illustrative.

1. Project brief

Background

Northwind Freight is a mid-sized Australian logistics company with about 400 staff and three warehouses, in Sydney, Brisbane and Melbourne. Its warehouse management system (WMS) is 14 years old, has been out of vendor support since 2025, and runs on an on-premises server that is near the end of its life. Stock levels only update in an overnight batch, meaning customer service cannot see live stock, and picking errors are usually found when a customer complains.

Objective

Replace the legacy WMS with a supported, cloud-hosted WMS across all three warehouses within 11 months, without disrupting the November to December peak season.

Scope

In scope	Out of scope
Selecting and contracting a new WMS vendor Configuring receiving, put-away, picking, packing and dispatch workflows Migrating item, location and on-hand stock data Integrating with the existing ERP (orders, invoicing) and carrier booking New handheld scanners for floor staff Training, cutover and handover to business-as-usual support	Replacing the ERP A transport management system Warehouse automation such as conveyors or robotics Changes to physical warehouse layouts

Success criteria

Criterion	Measure	Target
All sites live	Sites operating on the new WMS	3 of 3 by end of Month 10
Pick accuracy	Correct lines picked as a share of lines picked	99.5% or better (baseline 97.8%)
Stock visibility	Delay between a stock movement and the system showing it	Under 5 minutes (baseline: overnight)
Business continuity	Unplanned downtime during each site cutover	Under 8 hours per site

Criterion	Measure	Target
Security	ASD Essential Eight maturity of the new system	Maturity Level Two before go-live
Budget	Spend against the approved budget	Within 10% tolerance

Indicative budget: $1.4 million including 15% contingency. Budget management is one of my development areas, so this plan sets a single tolerance figure and monthly cost reporting rather than a detailed cost model.

2. Schedule

Planned start is January 2027. The project runs over 11 months in eight stages. Most stages are sequential and end in a stage gate, where the steering committee decides whether the project continues. Stage 2 runs as two-week sprints, and Stage 3 runs alongside it.

Stage	Months	Approach	Key deliverables	Ends with
0. Initiation	Jan	Sequential	Business case, project charter, steering committee formed	Gate 1: business case approved
1. Requirements and vendor selection	Feb to Mar	Sequential	High-level requirements (including Essential Eight controls), tender, vendor evaluation, contract	Gate 2: contract signed
2. Configuration	Apr to Jun	Iterative: six two-week sprints	Configured workflows, reviewed with floor staff at every sprint review	Configuration accepted by the product owner
3. Data migration and integration	May to Jul	Sequential, parallel to Stage 2	Cleansed data, ERP and carrier integrations, two full migration rehearsals	Rehearsal sign-off
4. Testing	Jul	Sequential	System, integration and user acceptance testing; security assessment	Gate 3: go or no-go for pilot
5. Pilot	Aug	Sequential	Brisbane live, measured against the success criteria	Gate 4: go or no-go for rollout
6. Rollout	Sep to Oct	Sequential	Sydney, then Melbourne, live over weekend cutovers	Rollout complete
7. Hypercare and handover	Nov	Sequential	Four weeks of close support, handover to business-as-usual IT	Gate 5: project closure

Peak season rule: no site goes live after 31 October. If Gate 4 slips by more than three weeks, the steering committee decides whether to defer the Sydney and Melbourne rollouts to February rather than cut over during peak.

3. Risk register

Likelihood (L) and impact (I) are rated from 1 (low) to 5 (high). Rating is L × I: 15 or more is high, 8 to 14 is medium, and under 8 is low.

ID	Risk	L	I	Rating	Treatment	Owner
R1	Legacy data is inaccurate (duplicate items, wrong locations), so the new system starts with wrong stock	4	4	16	Data cleansing in Stage 3, two full migration rehearsals, stocktake before each cutover	Data lead
R2	Vendor misses configuration milestones because its team is shared across clients, pushing go-live towards peak	3	4	12	Milestone-based payments in the contract, a named vendor team, weekly progress review	Project manager
R3	Floor staff resist the new scanners and workflows, leading to workarounds and low pick accuracy	3	4	12	Floor staff at every sprint review from Stage 2, a champion at each site, training on real tasks before go-live	Warehouse managers
R4	The new system fails security requirements (MFA, admin privileges, tested backups) and go-live is blocked	3	4	12	Essential Eight requirements written into the tender; security assessment scheduled in Stage 4, not after it	IT security lead
R5	Scope creep, such as requests for automation or transport features	4	3	12	Out-of-scope list in the charter; changes go through change control at the steering committee	Project manager
R6	A site cutover overruns and orders cannot ship	2	5	10	Weekend cutovers, a tested rollback plan, and a manual picking fallback	Project manager
R7	Rollout slips into the November to December peak	2	5	10	Peak season rule; Gate 4 decides whether to defer	Steering committee
R8	ERP integration is more complex than estimated	3	3	9	Integration spike in the first sprint; contingency held for integration work	Integration lead
R9	Operations decision-makers miss sprint reviews, so configuration is built on assumptions	3	3	9	A product owner from operations with time formally allocated; review dates fixed at project start	Project sponsor
R10	Costs overrun from hardware price rises or extra vendor days	3	3	9	15% contingency, fixed-price hardware quote, monthly cost report to the sponsor	Sponsor and finance

4. Stakeholder map

The grid places each stakeholder group by its power over the project and its interest in the outcome.

	Low interest	**High interest**
High power	Keep satisfied CEO; Chief Financial Officer	Manage closely Chief Operating Officer (sponsor); steering committee; warehouse managers; WMS vendor
Low power	Monitor Customers (through account managers); carriers	Keep informed, and engage actively Floor staff and team leaders; customer service; business-as-usual IT support

Stakeholder	**What they care about**	**Engagement approach**
Chief Operating Officer (sponsor)	Business case, peak season, cost	Chairs the steering committee; decides at each stage gate; monthly cost and risk report
Warehouse managers	Throughput and staff disruption	Steering committee members; approve the cutover plan for their own site
Floor staff and team leaders	Whether the new scanners and screens make their work easier or harder	Sprint reviews held on the warehouse floor; a champion at each site; training on real tasks
WMS vendor	Clear requirements, on-time payment	Weekly delivery meeting; milestone-based contract; named team
Customer service	Live stock visibility for customers	Represented in the backlog; early demo of stock enquiry screens
Business-as-usual IT support	A system they can support after handover	Involved from Stage 3; joint hypercare; formal handover at Gate 5
Chief Financial Officer	Spend against budget	Monthly cost report; consulted on any change above tolerance

Why floor staff are treated differently: on a standard grid, floor staff fall into "keep informed", which would mean training them at the end. They have the most influence on whether the system succeeds, because pick accuracy depends on how they use it every day. This plan treats them as co-designers through sprint reviews from Stage 2. That follows a lesson from my Goal 1 work: design with the people who will live with a system, not just for them.

5. Methodology justification

Decision: a hybrid approach. PRINCE2-style stages and gates govern the project as a whole, and Scrum is used inside the configuration stage.

Options considered

Approach	Where it fits this project	Where it does not fit
Waterfall or PRINCE2 only	Procurement, the vendor contract, data migration and cutover all need fixed sequences and formal go or no-go decisions.	Configuration. Requirements would be fixed before floor staff had used anything, so problems would only show up in acceptance testing in Month 7, when they are expensive to fix.
Scrum only	Configuring workflows and screens, where frequent feedback from floor staff is valuable.	A fixed-price vendor contract, a one-weekend cutover and a peak season deadline cannot be sprinted. Scrum also has no built-in point to ask whether the business case is still valid.
Hybrid (chosen)	Uses each approach where it fits: gates for the fixed, high-risk work, and sprints for the work that benefits from feedback.	Adds coordination effort between the two ways of working (see limitations below).

How the hybrid works

- The steering committee approves each stage at its gate. The project manager runs each stage within agreed tolerances (three weeks on time, 10% on cost) and escalates only when a tolerance is forecast to be exceeded. This is PRINCE2's management by exception.
- In Stage 2, a product owner from operations orders the configuration backlog. The developers, vendor consultants and Northwind IT staff, work in two-week sprints, with each sprint review held on a warehouse floor.
- The sprint backlog can change from sprint to sprint. The stage end date and budget cannot change without a decision at the next gate. That boundary is what keeps the two approaches from conflicting.

Where the project manager sits

Scrum does not define a project manager role, and inside Stage 2 the project manager does not direct the sprint work. A hybrid project still needs someone who owns the vendor contract, the budget, the risk register and the stage gates. That is the role I am working towards, and seeing that Scrum and project management can coexist was one of the main things I took from the Scrum Open assessment.

Limitations

- Two vocabularies and two rhythms (stages and sprints) can confuse stakeholders. A single integrated plan and one reporting format to the steering committee reduce this.
- Teams can treat sprints as small waterfall phases. The product owner and sprint reviews with real users are what prevent that.
- The choice depends on the vendor being willing to work in sprints. That should be tested during vendor selection in Stage 1, not assumed.